

Assignment:

Identify your problem statement: **to predict the insurance charges**

Tell basic info about the dataset (Total number of rows, columns): 1338 rows & 6 Columns

1. Domain Selection: Machine Learning
2. Learning Selection: Supervised Learning
3. Algorithm: Regression

age	sex	bmi	children	smoker	charges
19	female	27.9	0	yes	16884.924
18	male	33.77	1	no	1725.5523
28	male	33	3	no	4449.462
33	male	22.705	0	no	21984.47061

Gender & Smoker is in Nominal Data & need to convert in numbers (0,1)

Multi Linear Regression:

r_score = 0.7894790349867009

SVR: Support Vector Regression:

SVR	
rbf	- 0.08842732776913875
linear	-0.11166128719608448
poly	-0.06429258402105531
sigmoid	-0.0899412170256757
precomputed	NA

Rbf, C= 1.0 -> - 0.08842732776913875

Rbf, C= 0.1 -> - 0.08957624598812952

Rbf, c=10 => -0.08196910396420853

Rbf, c=100 -> -0.12480367775039669

DecisionTree:

<i>criterion='squared_error', splitter='best'</i>	0.709745022
criterion='friedman_mse', splitter='best'	0.714282831
criterion='absolute_error', splitter='best'	0.666568596
criterion='poisson', splitter='best'	0.728928924
<i>criterion='squared_error', splitter='random'</i>	0.700334648
criterion='friedman_mse', splitter='random'	0.764850174
criterion='absolute_error', splitter='random'	0.717433303
criterion='poisson', splitter='random'	0.705281106

Random Forest:

n_estimators=50,random_state=0	0.849832932
n_estimators=100,random_state=0	0.853830791
n_estimators=100,random_state=0,criterion='absolute_error'	0.852009362
n_estimators=100,random_state=0,criterion='friedman_mse'	0.854051894
n_estimators=100,random_state=0,criterion='poisson'	0.852633426

final model : Random Forest - ***friedman_mse***

R_score = 0.854051894